

DDA3020 Machine Learning

Final Exam

2022.12

1 Single-choice questions (2 points per question)

- 1.1 Which one is not a sub-area of artificial intelligence?
- A. Computer Vision
 - B. Machine Learning
 - C. Optimization
 - D. Natural Language Processing
- 1.2 Which one is not a continuous distribution?
- A. Gaussian distribution
 - B. Binomial distribution
 - C. Laplace distribution
 - D. Beta distribution
- 1.3 Which of the following would you apply unsupervised learning to?
- A. Compress a high-resolution image to a low-resolution image.
 - B. Develop a model to predict the stock market.
 - C. Given the data of body dimensions collected from 1,000 consumers, determine the size of the clothes to be produced.
 - D. Given a large dataset of medical records of patients including the disease name, predict the disease of the new patient by the symptom.
- 1.4 Which one is a non-parametric method?
- A. Linear regression with nonlinear basis functions
 - B. K-Nearest Neighbors classifier
 - C. Linear SVM
 - D. Neural network
- 1.5 Which one is incorrect about the bias-variance decomposition?
- A. When the bias increases, the variance decreases
 - B. When the model is more complex, the bias often becomes smaller
 - C. When the model is more complex, the variance often becomes larger
 - D. We can reduce the noise term
- 1.6 Which strategy can reduce the overfitting of a parametric model?
- A. Decrease the number of training samples.
 - B. Add regularization terms to the loss function
 - C. Consider a complex hypothesis model
 - D. Use a second-order optimization algorithm instead of gradient descent or SGD
- 1.7 Which statement about neural networks is correct?
- A. A neural network with only one nonlinear hidden layer does not have the universal approximation ability.
 - B. The size of the output cannot be larger than the size of the input.
 - C. A neural network without nonlinear activation functions is equivalent to a linear model.
 - D. CNN cannot contain a fully connected layer.
- 1.8 Which statement about decision tree is correct?
- A. Decision tree can only handle training data with numerical attributes.
 - B. Decision tree is a linear model.
 - C. When we set the minimal size of the leaf node to be a large value, then we will grow a deep decision tree.
 - D. When we set the maximal depth to be a small value, then we will grow a shallow decision tree.
- 1.9 A class has 10 students. They received marks for their mid-term quiz as follows.

Student ID	01	02	03	04	05
Marks	90	50	66	71	82
Student ID	06	07	08	09	10
Marks	72	75	68	92	60

To group the students into two tutorial groups according to their marks, we use k-means. We pick student 02 as the initial centroid for Group A, and 07 for Group B, and assign the students to the two groups using Euclidean distance. Which one of the following results is correct?

- A. We will have 4 students in Group A.
- B. We will have 5 students in Group B.
- C. If we change the initial centroid, the clustering result by kmeans will not be changed.
- D. With the above group assignments, we re-estimate the new centroids for the two groups. The new centroid of Group A is 55.

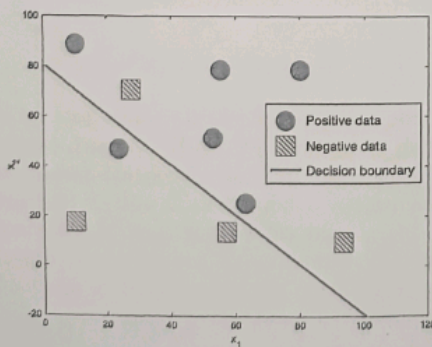
1.10 Which statement about PCA is incorrect?

- A. PCA is an unsupervised method.
- B. PCA can give non-linear projection of data.
- C. PCA chooses the k eigenvectors with the top k largest eigenvalues to form a new matrix.
- D. PCA can be used to reduce the noise of data.

1.11 Which one is incorrect for Gaussian mixture models (GMM)?

- A. GMM takes into account the mean and variance of the data.
- B. GMM can be solved by EM algorithm.
- C. GMM is a soft clustering method.
- D. The computational cost of GMM is lower than that of K-means.

1.12 In the following figure, the red circles represent the data points in the positive class and the blue squares represent the data points in the negative class. We construct a model to do the prediction and get a decision boundary as the black curve. The points on the top right of the decision boundary are predicted to be positive and other points at the bottom left of the boundary are predicted to be negative.



Which statement is incorrect?

- A. The precision in the figure is $\frac{5}{7}$.
- B. The recall in the figure is $\frac{4}{6}$.
- C. The accuracy is $\frac{7}{10}$.
- D. The higher AUC we get, the better the model is.

1.13 Given a linear system $Xw = y$, where $w \in \mathbb{R}^d$ is the variable we want to solve, and $X \in \mathbb{R}^{m \times d}$ and $y \in \mathbb{R}^m$ are the data provided, which of the following statement is correct?

- A. When $m = d$, it is guaranteed to obtain a unique solution even when $X^T X$ is not invertible.
- B. When $m > d$, this linear system is called under-determined system, and there is no solution.
- C. When $m > d$, this linear system is called over-determined system, and there is no solution.
- D. When $m < d$, this linear system is called under-determined system, and there is no solution.

1.14 Which statement is correct?

- A. Logistic regression maximizes the margin between two classes.
- B. The decision boundary given by softmax regression is nonlinear.
- C. The derivative of sigmoid function $g(a) = \frac{1}{1+\exp(-a)}$ is $\frac{dg}{da} = g(a)(1+g(a))$.
- D. Both logistic regression and softmax regression have no closed-form solutions.

1.15 Suppose we have 10 models from 10 different datasets. For a new test sample $(x, y) = (4, 9)$ sampled from the same distribution that generated the training sets, we suppose $t(x = 5) = 8.5$ (ground truth), and ϵ (noise) is instantiated as 0.5. Suppose the predictions for this new test sample given by the 10 trained models are 8, 7, 11, 6, 5, 14, 10, 9, 11, 6, respectively. Based on these 10 trials, which of the following results is incorrect?

- A. The empirical mean squared error (MSE) is 7.3.
- B. The bias² is 0.04.
- C. The variance is 7.1.

2 Calculations and Derivations (70 points)

2.1 (5 points) Consider two discrete random variables X and Y , $X \in \{0, 1, 2\}$, $Y \in \{2, 3\}$. $P(Y = 2) = 0.3$, $P(Y = 3) = 0.7$. Given Y , X follows binomial distribution, $P(X = k|Y = y) = \frac{n!}{k!(n-k)!} (\frac{1}{y})^k (1 - \frac{1}{y})^{n-k}$, where $n = 2$.

1. Calculate the probability distribution of X , i.e., $P(X)$. (1 point)
2. Calculate the conditional distribution of $P(Y|X = 1)$. (Hint: using Bayes rule) (2 points)
3. Calculate $E(X)$ and $E(Y|X = 1)$. (2 points)

2.2 (15 points) We define a CNN model as

$$f_{CNN}(X) = \text{Softmax}(FC_1(\text{Conv}_2(\text{MP}_1(\text{Relu}_1(\text{Conv}_1(X)))))$$

The size of the input data X is $28 \times 28 \times 3$; the first convolutional layer Conv_1 includes $10 \times 6 \times 6 \times 3$ filters,

stride=2, padding=1; $Relu_1$ indicates the first Relu layer; MP_1 is a 2×2 max pooling layer, stride=2; the second convolutional layer $Conv_2$ includes $50 \times 5 \times 5 \times 10$ filters, stride=1, padding=0; FC_1 indicates the fully connected layer, where there are 10 output neurons; $Softmax$ denotes the Softmax activation function. The ground-truth label of X is denoted as t , and the loss function used for training this CNN model is denoted as $\mathcal{L}(y, t)$.

1. Compute the feature map sizes after $Relu_1$ and $Conv_2$ (2 points)
2. Calculate the number of parameters of this CNN model (Hint: do not forget the bias parameter the in convolution layers and fully connected layer) (4 points)
3. Plot the computational graph (CG) of the forward pass of this CNN model (3 points) (Hint: use $z_1, z_2, z_3, z_4, z_5, z_6$ to denote the activated value after $Conv_1, Relu_1, MP_1, Conv_2, FC_1, Softmax$)
4. Based on the plotted CG, write down the formulations of the back-propagation algorithm, including the forward and backward pass (6 points) (Hint: for the forward pass, write down the process of how to get the value of loss function $\mathcal{L}(y, t)$; for the backward pass, write down the process of computing the partial derivative of each parameter, like $\frac{\partial \mathcal{L}}{\partial w_1}, \frac{\partial \mathcal{L}}{\partial b_1}$)

2.3 (17 points) Derivations of SVM's objective function and solution according to the dual problem. Given the training data $\{(x_i, y_i)\}_i^n$, denoting the parameters as $\{w, b\}$, the objective function of SVM is formulated as follows:

$$\min_{w, b} \frac{1}{2} \|w\|^2 \quad (1)$$

$$\text{s.t. } y_i (w^T x_i + b) \geq 1, i = 1, \dots, n.$$

1. Derive the above objective from the perspective of large margin. (Hint: The margin is defined as the closest distance from the data point to the hyperplane. We wish to find a hyperplane that separate the data, meanwhile having the largest margin among all the hyperplanes. The distance of a point x' to a hyperplane given by $f_{w, b}(x) := w^T x + b = 0$ is given by $\frac{|f_{w, b}(x')|}{\|w\|}$) (6 points)
2. Derive the above objective from the perspective of hinge loss. (Hint: The objective function of SVM using hinge loss is given by

$$C \sum_i \text{hinge_loss}(x, y; w, b) + \frac{1}{2} \|w\|^2$$

You can firstly write down the explicit expression of hinge loss.) (5 points)

3. Derive the solution of w, b according to the Lagrangian function and KKT condition, and tell

when a training data is called a support vector. (Hint: first write down the Lagrangian function, then KKT conditions, then use the KKT conditions to derive the solution.) (6 points)

2.4 (15 points) Suppose that a random variable x follows the Gaussian mixture distribution with $p(x) = \sum_{k=1}^K \pi_k \mathcal{N}(x | \mu_k, \Sigma_k)$. Where $\Theta = \{\pi, \mu, \Sigma\}$, $\pi = \{\pi_1, \dots, \pi_K\}$, $\mu = \{\mu_1, \dots, \mu_K\}$, $\Sigma = \{\Sigma_1, \dots, \Sigma_K\}$ and latent variable $z \sim \text{Categorical}(\pi)$, where $\pi \geq 0$, $\sum_k \pi_k = 1$.

1. Likelihood Decomposition. Show that

$$\ln p(\mathcal{D}; \Theta) \geq \mathcal{L}(q; \Theta), \forall q, \Theta,$$

where

$$\mathcal{L}(q; \Theta) = \sum_{n=1}^N \mathbb{E}_{q_n(z^{(n)})} \left[\ln \left(\frac{p^{(n)}(z^{(n)})}{q_n(z^{(n)})} \right) \right],$$

and $q(z) = \prod_{n=1}^N q_n(z^{(n)})$. And, write down the gap between $\ln p(\mathcal{D}; \Theta)$ and $\mathcal{L}(q; \Theta)$. (6 points)

2. E-Step Derivation. With given $\Theta = \{\pi, \mu, \Sigma\}$, update $q(z)$. (3 points)
3. M-Step Derivation Given $q(z)$, update $\Theta = \{\pi, \mu, \Sigma\}$. (6 points)

2.5 (8 points) Consider a binary classification with the following training data. Each data is described by 3 attributes: Color, Shape, Size.

Example	Color	Shape	Size	Class
1	Red	Square	Big	+
2	Blue	Square	Big	+
3	Red	Square	Small	+
4	Red	Circle	Big	+
5	Green	Circle	Big	-
6	Blue	Triangle	Small	-
7	Blue	Circle	Small	-
8	Green	Square	Big	-

1. Use the entropy method to find the best attribute for the root node. (Hint: Calculate the entropy and information gain) (4 points)
2. Build a complete tree according to the entropy method. The tree should include the class label and training data in each node. (2 points)
3. Given two test data "Red, Triangle, Big" and "Green, Square, Small", find their labels using to your tree.

2.6 (10 points) Given the training data $\mathcal{D} = \{x^{(1)}, \dots, x^{(N)}\}$, PCA finds an orthogonal matrix $U \in \mathbb{R}^{D \times K}$ to get lower-dimensional data $z_n = U^T(x_n - \mu)$, $n = 1, \dots, N$, where $\mu = \frac{1}{N} \sum_{n=1}^N x^{(n)}$. The reconstructed data are denoted by $\tilde{\mathcal{D}} = \{\tilde{x}^{(1)}, \dots, \tilde{x}^{(N)}\}$,

where $\tilde{\mathbf{x}}_n = \mathbf{U}\mathbf{z}_n + \boldsymbol{\mu}$. PCA has two formulations. The first one is the maximum variance formulation, i.e.,

$$\max_{\mathbf{U}^T \mathbf{U} = \mathbf{I}} \frac{1}{N} \sum_{n=1}^N \|\tilde{\mathbf{x}}^{(n)} - \tilde{\boldsymbol{\mu}}\|^2, \quad (2)$$

where $\tilde{\boldsymbol{\mu}} = \frac{1}{N} \sum_{n=1}^N \tilde{\mathbf{x}}^{(n)}$ denotes the mean of the reconstructions. The second one is the minimum reconstruction error formulation, i.e.,

$$\min_{\mathbf{U}^T \mathbf{U} = \mathbf{I}} \frac{1}{N} \sum_{n=1}^N \|\mathbf{x}^{(n)} - \tilde{\mathbf{x}}^{(n)}\|^2. \quad (3)$$

Prove that the two formulations are equivalent in terms of the solution of $\mathbf{U}$ and explain how to obtain $\mathbf{U}$ from the covariance matrix of the data.